VAMSHI KUMMARI

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PROFILE

AI/ML Engineer graduate with hands-on experience building and deploying Generative AI applications, multi-agent systems, and machine learning solutions. Skilled in LangChain, LangGraph, RAG pipelines, FastAPI, and cloud deployment, with a strong foundation in classical ML, deep learning, and NLP.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

GenAI / LLM Engineering: LangChain, LangGraph, RAG, Vector Databases (Qdrant), Embeddings, Prompt Engineering, LLM Evaluation

Machine Learning: Scikit-learn, Random Forest, XGBoost, LightGBM, Deep Learning (PyTorch), NLP

Backend & Deployment: FastAPI, REST APIs, Docker, Git/GitHub, Render, Streamlit

Data & Tools: Pandas, NumPy, Power BI, SQL, EDA

PROJECTS

ResearchMind AI – Multi-Agent Research Assistant

LangGraph | FastAPI | React/TypeScript | Groq (LLaMA-3.3-70B) | Mistral | Tavily | Live: researchmind-np15.onrender.com

- Built a multi-agent research platform that turns a user query into a structured, citation-backed report using a 5-stage LangGraph pipeline (Router → Research → Orchestrator → Workers → Reducer).
- Designed fan-out parallel execution so multiple Worker agents research and draft report sections concurrently, reducing total report generation time.
- Added configurable target audience and report-type settings so users can tailor report depth and style per query.
- Integrated the Tavily API for live web retrieval to ground responses in real sources and reduce hallucinations.
- Implemented real-time token streaming end-to-end via FastAPI SSE (astream_events) and a React ReadableStream reader, and deployed the full-stack app on Render.
- Built an LLM-as-judge evaluation harness across 24 test queries measuring faithfulness, citation accuracy, and hallucination risk, using findings to target citation-grounding improvements in report generation.

AI Mock Interviewer – Adaptive Interview Simulation Platform

LangChain (LCEL) | FastAPI | Groq LLaMA-3.3-70B | JavaScript | Docker | Live: mock-interview-7bui.onrender.com

- Built a 4-chain LCEL pipeline (question generation → evaluation → ideal answer → final report) covering 14+ topics from Python and DSA to System Design and HR/Behavioral.
- Supported Topic-Based, Job Role-Based, and Resume-Based interview modes with MCQ, written, and voice-based answer formats.
- Used Pydantic structured outputs to enforce consistent, parseable evaluation and feedback from the LLM.
- Fixed a subtopic-repetition bug by implementing a server-side subtopic cursor, ensuring non-repeating adaptive follow-up questions across a session.
- Added downloadable performance reports with Chart.js score trends; containerized with Docker and deployed on Render and Vercel.

LandSense AI – Land Price Prediction System

Python | Scikit-learn | Random Forest | XGBoost | LightGBM | Streamlit | Leaflet.js | Live: landsense-ai.streamlit.app

- Built a B2B land price prediction system trained on 65,000+ property records across 367 localities in 14 Indian states.
- Engineered 27 features (infrastructure quality, zoning, historical growth trends) and benchmarked Random Forest, XGBoost, and LightGBM models.
- Achieved 93% R^2 with an optimized Random Forest model, tuned via GridSearchCV.
- Deployed an interactive Streamlit + Leaflet.js app on Streamlit Cloud, using Git LFS to serve the 235MB model file.

EDUCATION

B.Tech, Computer Science & Engineering (Data Science) — Swami Vivekananda Institute of Technology, Hyderabad | 2021 – 2025